

Assessment Co1- At-1

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Question 1 Solution (5 Marks)

Question:

A parking facility has three sections containing cars of different colors. Section A has 5 red, 3 blue, and 2 black cars; Section B has 4 red, 6 blue, and 5 black cars; and Section C has 7 red, 2 blue, and 6 black cars. A section is chosen at random, but Section B is twice as likely to be selected as each of the other sections due to higher usage. A randomly selected car from the chosen section turns out to be blue. What is the probability that the car came from Section C?

Solution

Given:

Section A: Red=5, Blue=3, Black=2, Total=10

Section B: Red=4, Blue=6, Black=5, Total=15

Section C: Red=7, Blue=2, Black=6, Total=15

Since Section B is twice as likely to be selected as Sections A and C:

$P(A)=1/4$, $P(B)=1/2$, $P(C)=1/4$

Step 1: Probability of selecting a blue car from each section:

$P(\text{Blue}|A)=3/10$

$P(\text{Blue}|B)=6/15=2/5$

$P(\text{Blue}|C)=2/15$

Step 2: Total probability of selecting a blue car:

$P(\text{Blue})=(1/4 \times 3/10)+(1/2 \times 2/5)+(1/4 \times 2/15)$

$=3/40+1/5+1/30$

$=37/120$

Step 3: Using Bayes' Theorem:

$P(C|\text{Blue})=[P(C) \times P(\text{Blue}|C)]/P(\text{Blue})$

$=[(1/4) \times (2/15)]/(37/120)$

$=4/37 \approx 0.1081 = 10.81\%$

Final Answer

Probability that the blue car came from Section C = $4/37 \approx 10.81\%$.

Question 2: Training Loss and Validation Loss Analysis (5 Marks)

Given Data

Epoch	Training Loss	Validation Loss
1	2.8	2.4
2	2.5	2.2
3	2.2	2.0
4	2.0	2.1
5	1.8	2.3
6	1.6	2.6
7	1.5	2.9
8	1.3	3.2

Solution

i) Identify the epoch at which the model begins to overfit:

Training loss continuously decreases from Epoch 1 to Epoch 8, while validation loss decreases only until Epoch 3 and starts increasing from Epoch 4. Therefore, the model begins to overfit at Epoch 4.

ii) Two effective techniques to reduce overfitting:

1. Early Stopping – Stop training when the validation loss starts increasing.
2. Regularization (Dropout or L2 Regularization) – Reduces model complexity and improves generalization.

iii) Effect of overfitting on generalization:

Overfitting causes the model to memorize the training data instead of learning general patterns. As a result, the model performs well on the training data but poorly on unseen or new data, reducing its generalization ability.

Final Answers

i) Overfitting begins at Epoch 4.

ii) Techniques: Early Stopping and Regularization (Dropout/L2).

iii) Overfitting reduces the model's ability to generalize to unseen data.

Question 3: Training and Validation Accuracy Analysis (5 Marks)

Given Data

Epoch	Training Accuracy	Validation Accuracy
1	60%	58%
2	65%	63%
3	70%	68%
4	75%	72%
5	80%	73%
6	85%	72%
7	88%	70%
8	92%	68%

Solution

i) Identify the epoch at which the model starts overfitting:

Validation accuracy increases up to Epoch 5 (73%). From Epoch 6 onward, training accuracy continues to increase while validation accuracy decreases. Therefore, the model starts overfitting at Epoch 6.

ii) Why does validation accuracy decrease despite training accuracy increasing?

The model begins to memorize the training data instead of learning general patterns. As a result, it performs better on the training data but worse on unseen validation data, causing validation accuracy to decrease.

iii) Two methods to improve validation performance:

1. Early Stopping – Stop training when validation accuracy stops improving.
2. Regularization (Dropout or L2 Regularization) – Reduces overfitting and improves generalization.

Final Answers

i) Overfitting starts at Epoch 6.

ii) Validation accuracy decreases because the model memorizes the training data and cannot generalize well to unseen data.

iii) Methods: Early Stopping and Regularization (Dropout/L2).

